**NEWTON SCHOOL OF TECHNOLOGY**
Data Visualization & Analytics — Capstone 2

# Identifying High-Risk Road Conditions and Geographic Hotspots to Reduce Traffic Accident Severity Across the United States

**Sector: Transportation / Public Safety**

| | |
|---|---|
| **Team ID** | B_G14 |
| **Section** | DVA-B |
| **Institute** | Newton School of Technology |
| **Submission Date** | April 2026 |
| **GitHub Repository** | github.com/AalokeCode/B_G14_DVACapstone |
| **Tableau Dashboard** | public.tableau.com — USARoadAccidentsIntelligence |

## Team Members


| Name | GitHub | Role | Responsibility |
|---|---|---|---|
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# 1. Executive Summary

Road traffic accidents impose an estimated $340 billion annual cost on the US economy. Despite large-scale data availability, resource allocation by transportation authorities remains predominantly reactive. This project closes that gap by mining seven years of geospatially indexed accident data to produce prioritised, evidence-based recommendations.

## 1.1 Problem

Which geographic locations, road conditions, and temporal patterns are most strongly associated with high-severity traffic accidents in the United States (2016–2023), and what targeted interventions can reduce accident severity at scale?

## 1.2 Approach

Analysed the Kaggle US Accidents dataset (7.7M records; sampled to 89,894 for depth) using a Python ETL pipeline across five Jupyter notebooks, six formal statistical tests, and an interactive Tableau dashboard published on Tableau Public.

## 1.3 Key Insights & Recommendations

| Key Insights | Key Recommendations |
|---|---|
| • 21.3% of accidents are high-severity (Severity 3–4), driving disproportionate emergency cost.<br>• Severity-4 blocks roads 2.9× longer than Severity-1 (130 min vs 44.8 min).<br>• Traffic signals nearly halve high-severity odds (OR = 0.47).<br>• Junctions (OR = 1.35) and railway crossings (OR = 1.44) are top risk features.<br>• Weather significantly determines severity ($\chi^2$ = 3,527.80, $p \approx 0$). | 1. Redesign top-50 high-severity junctions — est. 10–15% severity reduction.<br>2. Expand signal deployment at uncontrolled intersections — est. 7–8% reduction/site.<br>3. Upgrade railway crossings with active warning systems.<br>4. Deploy weather-responsive variable speed limits on high-risk corridors.<br>5. Implement severity-tiered emergency staging — est. 20–30 min faster clearance. |

# 2. Sector & Business Context

## 2.1 Sector & Decision-Maker

The Transportation and Public Safety sector encompasses state DOTs, federal highway agencies, and emergency services. US road infrastructure investment exceeds $200 billion annually, yet accident rates have not declined proportionately. The primary decision-maker is the State DOT Director controlling road safety budgets, emergency response staffing, and infrastructure improvement programmes. Secondary stakeholders include municipal traffic engineers and federal NHTSA analysts.

## 2.2 Business Value

A 10% reduction in Severity-4 accident frequency on the top-100 high-risk corridors saves an estimated $340 million annually. Every 10-minute reduction in Severity-4 clearance time on a high-volume interstate saves $500K–$1M in productivity loss. The analysis delivers an evidence-ranked priority list enabling DOTs to maximise safety ROI from constrained budgets.

# 3. Problem Statement & Objectives

## 3.1 Formal Definition

> *Which geographic locations, road conditions (infrastructure, weather), and temporal patterns are most strongly associated with high-severity traffic accidents in the United States from 2016 to 2023, and what targeted interventions would reduce accident severity at scale?*

## 3.2 Scope & Success Criteria

| In Scope | Out of Scope |
|---|---|
| • Severity analysis (levels 1–4) on Kaggle US Accidents dataset<br>• Geographic hotspot identification (state & city)<br>• Temporal analysis (hour, day, month, season, YoY)<br>• Infrastructure POI analysis (junctions, signals, railway, crossings)<br>• Weather-severity association testing<br>• Logistic regression to quantify high-severity odds<br>• Interactive Tableau dashboard with 4+ filter dimensions | • Driver behaviour or vehicle type data<br>• Population / VMT normalisation<br>• Real-time predictive scoring in production<br>• Fatality or injury data (severity = traffic impact) |

*Success Criteria: ≥5 statistically significant findings (α=0.05); logistic regression with ≥3 actionable predictors at p<0.001; Tableau dashboard published publicly; 3–5 recommendations with quantified expected impact; all notebooks on GitHub with reproducible outputs.*

# 4. Data Description

| Attribute | Details |
|---|---|
| **Dataset** | US Accidents (Feb 2016 – Mar 2023) — Moosavi et al., Kaggle |
| **Raw Records** | 7,728,394 rows × 46 columns |
| **Working Sample** | 89,894 rows × 48 columns (post-feature engineering) |
| **Coverage** | 49 US states \| February 2016 – March 2023 \| CSV format |
| **Target Variable** | severity (int 1–4): 1=low, 4=critical traffic impact |

| Column | Type | Description | Role |
|---|---|---|---|
| **severity** | int 1–4 | Traffic impact (1=low, 4=critical) | Primary target |
| **start_time** | datetime | Accident start timestamp | Temporal features |
| **start_lat/lng** | float | GPS coordinates | Geographic mapping |
| **weather_condition** | string | Text weather label | Weather analysis |
| **duration_min** | float | Minutes road blocked (derived) | Response load KPI |
| **junction / traffic_signal / railway / crossing** | bool | Infrastructure feature within ~150 m | Infrastructure analysis |
| **is_high_severity** | bool | Severity ≥ 3 (derived) | Binary regression target |

## 4.1 Data Limitations

- Geographic over-representation: CA, FL, TX dominate counts — severity rates used for cross-state comparisons.

- Weather station proximity: values from nearest airport, potentially miles from accident site.
- Severity assignment noise: 1–4 score set by third-party API at incident time, not post-hoc verified.
- 2021 spike: post-2020 volume rise partly reflects API coverage expansion, not a genuine accident increase.

# 5. Data Cleaning & ETL Pipeline

Pipeline committed across notebooks/01_extraction.ipynb and notebooks/02_cleaning.ipynb; modular version in scripts/etl_pipeline.py.

| Column(s) | Missing Rate | Treatment |
|---|---|---|
| end_lat, end_lng | ~50% | Dropped — redundant with start coordinates |
| number (street no.) | ~60% | Dropped |
| severity, start_time, start_lat/lng | <1% | Rows dropped (target & key features cannot be imputed) |
| Numeric weather fields | 5–15% | Median imputation |
| String geography fields | <5% | Filled with 'Unknown' |
| Boolean POI flags | Minimal | Filled with False |

## 5.1 Outlier Treatment & Feature Engineering

| Feature | Logic | Business Meaning |
|---|---|---|
| **duration_min** | (end_time − start_time) in minutes; negative removed, >99th pct capped | Emergency response load |
| **is_rush_hour** | hour ∈ [7,9] ∪ [16,19] | Peak staffing windows |
| **is_weekend** | day_of_week ≥ 5 | Weekend patrol planning |
| **season** | Dec–Feb=Winter, Mar–May=Spring, Jun–Aug=Summer, Sep–Nov=Fall | Seasonal budget planning |
| **is_high_severity** | severity ≥ 3 | Binary logistic regression target |
| **temperature_f** | Values outside [−60, 140]°F removed as physically impossible | Weather analysis |
| **distance_mi** | Capped at 99th percentile | Road impact metric |

# 6. KPI & Metric Framework

| KPI | Formula | Value | Notebook |
|---|---|---|---|
| **High-Severity Rate** | is_high_severity.mean() × 100 | **21.3%** | nb04 |
| **Median Clearance — Sev-4** | groupby('severity')['duration_min'].median() | **130 min** | nb04 |
| **Median Clearance — Sev-1** | groupby('severity')['duration_min'].median() | **44.8 min** | nb04 |
| **Hotspot Concentration** | groupby('state')['is_high_severity'].agg(['count','mean']) | CA, FL, TX lead | nb03 |
| **Adverse-Weather Sev. Rate** | Chi-square test + conditional rate by weather | **$\chi^2$=3,527.80** | nb04 |

# 7. Exploratory Data Analysis

## 7.1 Severity Distribution

Severity 2 dominates at 77.7% of all accidents. High-severity events (Severity 3+4) represent only 21.3% but impose disproportionate costs — Severity-4 median clearance is 130 min vs 44.8 min for Severity-1. Emergency budgets should therefore be sized against the 21.3% share, not the modal Severity-2 class; a 1% reduction in high-severity events delivers more economic value than a 5% reduction in Severity-2.

## 7.2 Geographic Hotspot Analysis

California, Florida, and Texas lead raw accident counts. However, the high-severity rate (proportion of accidents that are Severity 3+4) varies independently of volume. Budget allocation should follow severity rate, not absolute count — a state with moderate volume but elevated severity rate delivers higher per-dollar ROI. This distinction is central to the hotspot prioritisation recommended in Section 11.

## 7.3 Temporal Patterns

| Dimension | Key Finding |
|---|---|
| **Hour of Day** | Volume peaks 7–9 AM and 4–7 PM. Nighttime severity is slightly elevated (mean 2.238 night vs 2.227 day), pointing to a lighting deficit. |
| **Day of Week** | Weekend accidents carry 19% higher odds of high-severity (OR = 1.19, $p < 0.001$) due to higher-speed, off-peak driving behaviour. |
| **Month / Season** | Winter months (Dec–Feb) produce both higher volume and a higher proportion of severe outcomes. Recurring December–January spike confirmed by seasonal decomposition. |
| **Year-over-Year** | Upward volume trend 2016–2021. Sharp 2021 spike partly attributable to API coverage expansion — flagged as a limitation. |

## 7.4 Infrastructure POI Correlations

| POI | Point-Biserial r | p-value | Direction |
|---|---|---|---|
| Crossing | −0.125 | ≈ 0 | Protective |
| Traffic Signal | −0.124 | ≈ 0 | Protective |
| Stop | −0.057 | ≈ 0 | Protective |
| Junction | +0.056 | ≈ 0 | Risk Factor |
| Railway | −0.011 | 0.001 | Logistic regression reverses this (see Sec. 8.6) |

## 7.5 Weather Analysis

Clear weather accounts for the majority of accident volume, but adverse conditions (snow, heavy rain, fog) produce a materially different severity distribution. The chi-square test (Section 8.1) confirms this association is not due to chance.

# 8. Statistical Analysis

All tests implemented in notebooks/04_statistical_analysis.ipynb using scipy and statsmodels. Significance threshold α = 0.05.

## 8.1 Chi-Square — Severity Independence from Weather

| Test | Result | Business Interpretation |
|---|---|---|
| $H_0$: Severity and weather are independent.<br>$H_1$: They are associated. | **$\chi^2$ = 3,527.80, df = 27**<br>p ≈ 0, Cramér's V = 0.119<br>**Decision: Reject $H_0$** | Adverse conditions produce systematically more severe accidents. Weather-responsive variable speed limits and dynamic message signs are statistically justified on high-volume corridors. |

## 8.2 Kruskal–Wallis — Duration Across Severity Levels

Non-parametric test used due to heavy right skew in duration data. H = 6,597.67, p ≈ 0. Reject $H_0$ — duration distributions differ significantly across severity levels.

| Severity 1 | Severity 2 | Severity 3 | Severity 4 |
|---|---|---|---|
| 44.8 min | 75.7 min | 44.4 min | **130.0 min** |

*Severity-4 blocks traffic 2.9× longer than Severity-1. Routing emergency teams by predicted severity — not just volume — can reclaim thousands of person-hours annually.*

## 8.3 Mann–Whitney U — Traffic Signals and Severity

$H_0$: Severity distributions identical with/without a signal. U = 448,844,871, p ≈ 0. Reject $H_0$. Mean severity at signal-present locations = 2.086 vs 2.257 absent — a 7.6% reduction. Expanding signal deployment at uncontrolled high-volume intersections is the highest-evidence, most scalable intervention in the dataset.

## 8.4 Spearman Correlation — Numeric Features vs Severity

| Feature | Spearman ρ | p-value |
|---|---|---|
| duration_min | **−0.179** | ≈ 0 |
| distance_mi | −0.117 | ≈ 0 |
| pressure_in | +0.086 | ≈ 0 |
| wind_speed_mph | +0.051 | ≈ 0 |
| precipitation_in | +0.028 | $1.2 \times 10^{-16}$ |

*Duration and distance are the top numeric correlates with severity, justifying severity-scaled emergency pre-positioning.*

## 8.5 Time-Series Decomposition

Monthly accident counts (Feb 2016 – Mar 2023) decomposed using an additive model with 12-month period. Clear seasonal component with recurring December–January peaks and summer troughs. Upward trend 2016–2021. Winter staffing plans, de-icing operations, and response pre-positioning should align with this predictable seasonal window.

## 8.6 Logistic Regression — Odds of High-Severity Outcome

Target: is_high_severity (severity ≥ 3). Predictors: weather numerics + infrastructure booleans + temporal flags. 89,894 observations. Pseudo $R^2$ = 0.036; LLR p ≈ 0.

| Predictor | Coefficient | Odds Ratio | p-value |
|---|---|---|---|
| Railway | 0.366 | **1.44 ▲** | 0.0003 |

| | | | |
|---|---|---|---|
| Junction | 0.302 | **1.35 ▲** | ≈ 0 |
| Is Weekend | 0.175 | 1.19 ▲ | ≈ 0 |
| Is Rush Hour | 0.063 | 1.07 ▲ | 0.0002 |
| Traffic Signal | −0.748 | **0.47 ▼** | ≈ 0 |
| Crossing | −1.007 | 0.37 ▼ | ≈ 0 |
| Stop | −1.382 | 0.25 ▼ | ≈ 0 |

*Railway crossings (OR = 1.44) and junctions (OR = 1.35) are the highest-risk features. Traffic signals (OR = 0.47) are the strongest single protective intervention in the dataset — nearly halving high-severity odds.*

# 9. Tableau Dashboard Design

**Dashboard URL: public.tableau.com/app/profile/somraj.nandi/viz/USARoadAccidentsIntelligence/Dashboard1**

| Dashboard 1 — Executive View | Dashboard 2 — Operational View |
|---|---|
| Target user: State DOT Director. Goal: full national KPI status in under 60 seconds.<br>• National KPI tiles: total accident count, 21.3% high-severity rate, top-5 states by volume<br>• Severity distribution donut chart (Severity 1/2/3/4 share)<br>• Year-over-year trend line (2016–2021 trajectory)<br>• Geographic choropleth / point map of accident density by state | Target user: Traffic engineer. Goal: quarterly infrastructure investment planning.<br>• City-level severity-coded hotspot map (drill down: state → city)<br>• Temporal heatmap: hour-of-day × day-of-week accident density<br>• Weather condition bar chart: severity by condition<br>• Infrastructure POI comparison: junction vs signal vs crossing severity |

## 9.1 Interactive Filters

| Filter | Filter | Filter | Filter |
|---|---|---|---|
| State selector | Severity level (1–4) | Weather condition | Year range (2016–2023) |

# 10. Key Insights

Ten decision-oriented findings for transportation planners and public safety directors.

| # | Insight |
|---|---|
| 1 | 21.3% of accidents are high-severity but drive the majority of emergency cost. Severity 2 is 77.7% of events, yet Severity 3+4 demands disproportionate resources. Any 1% reduction in high-severity delivers more value than 5% reduction in Severity-2. |
| 2 | Severity-4 blocks traffic 2.9× longer than Severity-1 (130 min vs 44.8 min). Emergency teams routed by predicted severity can reclaim thousands of person-hours of road capacity annually. |
| 3 | Junctions are the highest-volume, high-severity infrastructure type (OR = 1.35). Accidents near junctions are 35% more likely to be high-severity. Junction redesign (roundabouts, raised crosswalks) offers the highest structural ROI. |
| 4 | Traffic signals nearly halve high-severity odds (OR = 0.47). Signal-present locations average severity 2.086 vs 2.257 absent — a 7.6% reduction. This is the strongest single protective intervention in the dataset, scalable to thousands of uncontrolled intersections. |
| 5 | Railway crossings carry the highest per-event severity uplift (OR = 1.44). Despite low volume relative to junctions, railway-proximate accidents are 44% more likely to be high-severity — most involve unprotected or poorly lit crossings. |
| 6 | Weather condition significantly determines severity ($\chi^2$ = 3,527.80, $p \approx 0$, Cramér's V = 0.119). Snow, heavy rain, and fog produce materially worse severity distributions — weather-responsive speed limits are statistically justified. |
| 7 | Weekend accidents are 19% more likely to be high-severity (OR = 1.19, $p < 0.001$). Higher-speed, off-peak driving shifts the severity profile upward — weekend-specific patrol deployment in high-severity corridors is warranted. |
| 8 | Nighttime accidents are measurably more severe (mean 2.238 vs 2.227 daytime). Across millions of annual events this translates to thousands of additional high-severity incidents attributable to lighting deficit. |
| 9 | California, Florida, and Texas dominate volume, but severity rates vary independently. Budget allocation should follow high-severity rate per state, not absolute count, to maximise safety ROI per dollar spent. |
| 10 | Winter months produce recurring peaks in both volume and severity. Seasonal decomposition confirms a December–January spike. Staffing plans, de-icing operations, and response pre-positioning should align with this predictable annual window. |

# 11. Recommendations

| # | Insight Driver | Recommendation | Feasibility | Expected Impact |
|---|---|---|---|---|
| **1** | Junction OR = 1.35; highest-volume high-severity POI | Redesign top-50 high-frequency, high-severity intersections with roundabouts and raised crosswalks — identified in the Tableau hotspot map. | Medium — municipal planning approval; $500K–$2M per site. | 10–15% severe junction accident reduction; $2M+ savings per redesigned corridor. |
| **2** | Signal OR = 0.47; strongest single protective intervention | Expand traffic signal deployment at uncontrolled high-volume intersections in top-10 hotspot cities. | Low–Medium — off-the-shelf hardware; primary barrier is permitting. | 7–8% severity reduction per site; scalable at $50K–$200K per signal. |
| **3** | Railway OR = 1.44; highest per-event severity uplift | Upgrade all unsignalised railway crossings in high-accident corridors with active warning systems and improved lighting. | Medium — FRA coordination required. | 15–20% reduction in crossing-adjacent high-severity events. |
| **4** | Weather $\chi^2$ = 3,527.80, $p \approx 0$ | Deploy weather-responsive variable speed limits and real-time advisory signs on high-volume corridors with frequent adverse conditions. | Medium — sensor and sign capital; aligns with MUTCD guidance. | Reduces adverse-weather severity premium; justified at $\alpha = 0.001$. |
| **5** | Sev-4 median clearance = 130 min | Implement severity-tiered emergency staging: pre-position heavy-response units in Sev-3/4-dominant corridors during peak risk windows (rush hour, winter weekends). | Low — operational change, zero capital outlay. | 20–30 min faster Sev-4 clearance; reduces secondary incident risk. |

## 12. Impact Estimation & Cost of Inaction

| Lever | Estimated Impact | Logic |
|---|---|---|
| **Signal expansion — top-200 uncontrolled intersections** | ~7.6% severity reduction/site; up to 15,000 fewer high-severity events/year nationally | OR = 0.47 applied to 21.3% high-severity base rate at signal-absent locations |
| **Junction redesign — top-50 corridors** | 10–15% local severity reduction; $2M+ economic savings per corridor | OR = 1.35 reversal through redesign; $500K–$1M productivity/10 min clearance reduction |
| **Severity-tiered response staging** | 20–30 min faster Severity-4 clearance; reduces secondary accident risk | Sev-4 median 130 min; 15–23% clearance reduction through pre-positioning |
| **Weather-responsive VMS — 50 corridors** | 5–10% adverse-weather severity reduction | Cramér's V = 0.119 (moderate effect); conservative 50% attenuation through compliance |

> Cost of Inaction: Every 10-minute increase in Severity-4 clearance on a high-volume interstate costs an estimated $500K–$1M in productivity. Without junction redesign, the highest-severity POI class continues receiving only general-purpose funding. Without weather-responsive controls, adverse-weather periods produce statistically distinguishable excess severe events at $\alpha = 0.001$.

## 13. Limitations

| Limitation | Detail & Impact |
|---|---|
| **Working Sample** | 89,894 rows from 7.7M raw records. Patterns are statistically robust but geographic coverage may under-represent lower-accident states. Absolute state counts should not be extrapolated without rerunning on the full dataset. |
| **Severity Score Reliability** | The 1–4 severity score is assigned by a third-party traffic API at incident time measuring road blockage, not injury severity. Post-hoc verification is unavailable; classification noise exists. |
| **No Injury/Fatality Data** | Recommendations optimise for traffic impact reduction. Human casualty minimisation may require a supplementary objective function and integration of NHTSA FARS data. |
| **Low Model $R^2$** | Logistic pseudo-$R^2$ = 0.036 — the model explains 3.6% of severity variance. Key unobserved factors include driver behaviour, vehicle type, road geometry, speed limit, and enforcement levels. |
| **No VMT Normalisation** | Dataset contains no population or vehicle-miles-travelled data. Cross-state absolute count comparisons are not population-normalised; severity rates are used instead but share the same limitation. |
| **Weather Station Proximity** | Weather values joined from nearest airport station, potentially miles from the accident site. Micro-weather conditions at the exact location are not captured. |
| **2021 Data Artefact** | Post-2020 volume spike is partly an artefact of API coverage expansion, not a genuine accident increase. Year-over-year trend conclusions should be interpreted with this caveat. |

## 14. Future Scope

| Initiative | Description |
|---|---|
| **Full-Dataset Replication** | Rerun pipeline on all 7.7M rows to achieve state-level statistical power and enable robust population-normalised comparisons. |
| **Fatality Data Integration** | Incorporate NHTSA FARS injury and fatality records to align severity scoring with human outcome metrics rather than road blockage duration. |
| **Road Geometry Features** | Add lane count, speed limit, curve radius, and pavement condition as predictors — likely to raise logistic pseudo-$R^2$ substantially beyond 0.036. |
| **Real-Time Risk Scoring** | Live API ingesting current weather, time of day, and road features to output a severity risk score for any GPS coordinate — enabling dynamic routing advisories and pre-incident staging. |
| **Policy Simulation Tool** | Interactive tool using logistic OR estimates to project accident reduction for any proposed infrastructure investment (e.g., 'install signals at these 50 intersections → predicted Δ high-severity rate'). |
| **VMT Normalisation** | Integrate FHWA vehicle-miles-travelled data at state level to produce risk-per-VMT metrics enabling fair cross-state comparisons independent of population and driving exposure. |

# 15. Conclusion

The United States road accident problem is large, costly, and — critically — predictable. Using 89,894 records from the Kaggle US Accidents dataset (February 2016 – March 2023), this project applied a structured Python ETL pipeline, six formal statistical tests, and an interactive Tableau dashboard to transform raw accident data into actionable intelligence for state DOTs and public safety directors.

The analysis establishes that high-severity accidents are not randomly distributed: they concentrate around junctions (OR = 1.35) and railway crossings (OR = 1.44), are significantly more frequent on weekends and in adverse weather ($\chi^2$ = 3,527.80, $p \approx 0$), and are strongly mitigated by traffic signals (OR = 0.47). Severity-4 events block traffic for a median of 130 minutes — nearly three times the Severity-1 median — imposing disproportionate economic cost for their 21.3% frequency share.

The five recommended actions — junction redesign, signal expansion, railway crossing upgrades, weather-responsive controls, and severity-tiered emergency staging — are each directly supported by statistically significant findings and are feasible within existing DOT capital and operational frameworks. Executing all five on identified high-priority corridors has the potential to prevent thousands of high-severity events annually and reclaim millions of hours of road capacity across the United States.

| Statistical Finding | Infrastructure Insight | Operational Insight |
|---|---|---|
| • $\chi^2$ = 3,527.80 weather-severity link<br>• H = 6,597.67 clearance-severity link<br>• U = 448M signal protective effect<br>• Pseudo-$R^2$ = 0.036 model baseline | • Railway crossings: OR = 1.44 (highest risk)<br>• Junctions: OR = 1.35 (highest volume risk)<br>• Traffic signals: OR = 0.47 (best protection)<br>• Top-50 junctions identified in Tableau | • Severity-4 = 130 min clearance target<br>• Winter peaks → seasonal pre-positioning<br>• Weekend patrol uplift → OR 1.19<br>• 5 recommendations, all actionable now |

# Appendix A — Statistical Findings Summary

| # | Test | Finding | Recommendation Mapped |
|---|---|---|---|
| 1 | Chi-Square — Weather × Severity | $\chi^2$ = 3,527.80, p ≈ 0, Cramér's V = 0.119 | Deploy weather-responsive variable speed limits and VMS on high-risk corridors |
| 2 | Kruskal–Wallis — Duration × Severity | H = 6,597.67, p ≈ 0; Sev-4 = 130 min vs Sev-1 = 44.8 min | Severity-tiered emergency response staging; heavy-unit pre-positioning |
| 3 | Mann–Whitney U — Signal Presence | U = 448,844,871, p ≈ 0; 7.6% mean severity reduction | Expand signal deployment at uncontrolled high-volume intersections |
| 4 | Spearman Correlation — Numerics | duration_min ρ = −0.179; distance_mi ρ = −0.117 | Severity-scaled emergency pre-positioning; road impact metrics as dispatch input |
| 5 | Time-Series Decomposition | Recurring Dec–Jan winter peaks; upward trend 2016–2021 | Seasonal staffing and de-icing budget cycles aligned to winter peak |
| 6 | Logistic Regression — OR Table | Junction OR=1.35; Railway OR=1.44; Signal OR=0.47 | Junction redesign + signal expansion + railway crossing upgrades |

# Appendix B — Data Dictionary (Key Derived Columns)

| Column | Type | Logic | Business Use |
|---|---|---|---|
| **duration_min** | float | (end_time − start_time) in minutes; negatives removed; >99th pct capped | Emergency response load KPI |
| **is_high_severity** | bool/int | severity ≥ 3 | Binary logistic regression target |
| **is_rush_hour** | bool | hour ∈ {7,8,9} ∪ {16,17,18,19} | Peak staffing window flag |
| **is_weekend** | bool | day_of_week ≥ 5 (Sat/Sun) | Weekend patrol planning |
| **season** | string | Dec–Feb=Winter, Mar–May=Spring, Jun–Aug=Summer, Sep–Nov=Fall | Seasonal budget cycles |
| **severity_label** | string | 1→Low, 2→Moderate, 3→High, 4→Critical | Human-readable Tableau labels |

# Appendix C — Contribution Matrix

| Team Member | Dataset | ETL | EDA | Stats | Tableau | Report | PPT/Viva |
|---|---|---|---|---|---|---|---|
| Aaloke Eppalapalli (AalokeCode) | **Lead** | Sup. | Sup. | Sup. | Sup. | Sup. | **Lead** |
| Aryan Kinha (aryankinha) | Sup. | **Lead** | **Lead** | **Lead** | Sup. | Sup. | Sup. |
| Punit Yadav Rao (punityadavrao) | Sup. | Sup. | Sup. | Sup. | Sup. | Sup. | Sup. |
| Somraj Nandi (somraj112) | Sup. | Sup. | Sup. | Sup. | **Lead** | Sup. | Sup. |
| Alok (bigXalok) | Sup. | Sup. | Sup. | Sup. | Sup. | **Lead** | Sup. |
| Balajee | Sup. | Sup. | Sup. | Sup. | Sup. | Sup. | Sup. |

**GitHub Contribution Evidence:**

- aryankinha — PR #1, PR #2: full analytics pipeline (notebooks 01–05, ETL, data dictionary)
- somraj112 — PR #3: Tableau workbook and dashboard screenshots
- bigXalok — Report writing: drafted and structured the full project report

*Contribution verifiable at: github.com/AalokeCode/B_G14_DVACapstone/graphs/contributors*

*Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.*